

Lumen LF-60 Optical Flow Meter Datasheet

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Overview

The Lumen LF-60 is an inline optical-correlation flow meter for clean and lightly loaded liquids in general industrial, building services and water distribution applications. A pulsed LED light sheet passes through two sapphire windows in the measuring tube, and the electronics cross-correlate the particle and micro-bubble patterns seen by two receivers to calculate flow velocity. With no moving parts and no obstruction in the bore, the LF-60 produces negligible pressure loss and needs no routine recalibration.

The LF-60 is the standard-pressure member of the Lumen family. It is rated PN16 (16 bar) and suits pipe sizes from DN25 to DN200. For higher pressures, higher fluid temperatures, HART communication or IP67 installations, specify the LF-60P.

Key features

- Optical correlation measurement with a $\pm 1.0\%$ of reading accuracy across 0.5 to 12 m/s
- Full bore design with no moving parts and no pressure drop worth measuring
- 4-20 mA, scaled pulse and Modbus RTU (RS-485) outputs as standard
- Aluminium electronics housing with a 316L measuring tube and sapphire windows
- Two-line backlit display with local totaliser and diagnostics
- Works in water with turbidity as low as 0.2 NTU

Technical specifications

Parameter | LF-60 value

Nominal sizes | DN25 to DN200
Pressure rating | PN16 (16 bar)
Accuracy | $\pm 1.0\%$ of reading (0.5 to 12 m/s); ± 0.5 cm/s below 0.5 m/s
Velocity range | 0.1 to 12 m/s
Repeatability | $\pm 0.2\%$
Fluid temperature | -10 to +80 °C
Ambient temperature | -20 to +60 °C
Outputs | 4-20 mA, pulse (max 1 kHz), Modbus RTU
Power supply | 24 VDC $\pm 10\%$, 4 W
Ingress protection | IP65
Housing material | Powder-coated aluminium
Wetted materials | 316L stainless steel, sapphire windows, EPDM seals
Process connection | EN 1092-1 PN16 flanges (ASME Class 150 on request)
Weight (DN50) | 6.2 kg
Straight run required | 10D upstream, 5D downstream
Modbus defaults | Address 1, 9600 baud, 8N1
Current firmware | 1.8.2
Cable entries | 2 x M20 x 1.5

Ordering information

Order codes follow the pattern LF60-{size}-PN16, where the size is the three-digit nominal diameter. For example, LF60-050-PN16 is the DN50 meter and LF60-100-PN16 is the DN100 meter. Prices are published in the current regional price list.

Typical applications

- District heating and chilled water circuits below 80 °C
- Potable water distribution and pumping stations
- Cooling water monitoring on process plant
- Irrigation and treated effluent at low pressure

Installation notes

Install the LF-60 with at least 10 pipe diameters of straight pipe upstream and 5 downstream of the meter. The measuring tube must remain full; on vertical pipes, install with upward flow.

Full installation, wiring and flange bolting instructions are given in the Lumen LF-60 Installation Manual (prod-lf60-installation-manual).

Compliance

CE and UKCA marked. Pressure Equipment Directive 2014/68/EU, sound engineering practice for DN25 to DN50, category I for DN65 to DN200. EMC to EN 61326-1 industrial environment. The LF-60 is not certified for use in hazardous areas.